Learning of Uplink Resource Allocation with Multiuser QoS Constraints

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***Abstract*—In the paper the joint optimization of uplink multiuser power and resource block (RB) allocation are studied, where each user has quality of service (QoS) constraints on both long- and short-blocklength transmissions. The objective is to minimize the consumption of RBs for meeting the QoS requirements, leading to a mixed-integer nonlinear programming (MINLP) problem. We resort to deep learning to solve the problem with low inference complexity. To provide a performance benchmark for learning based methods, we propose a hierarchical algorithm to find the global optimal solution in the single-user scenario, which is then extended to the multiuser scenario. The design of the learning method, however, is challenging due to the discrete policy to be learned, which results in either vanishing or exploding gradient during neural network training. We introduce two types of smoothing functions to approximate the involved discretizing processes and propose a smoothing parameter adaption method. Another critical challenge lies in guaranteeing the QoS constraints. To address it, we design a nonlinear function to intensify the penalties for minor constraint violations. Simulation results demonstrate the advantages of the proposed method in reducing the number of occupied RBs and satisfying QoS constraints reliably.**

***Index Terms*—Deep learning, resource allocation, QoS constraints, MINLP.**

I. Introduction

The fifth-generation (5G) and beyond systems are designed to support diverse quality of service (QoS) requirements [1,2]. For uplink transmission, jointly optimizing the allocation of users' transmit power and the system's resource blocks (RBs) is crucial to guarantee the QoS requirements for both long-blocklength transmission (LBT) and short-blocklength transmission (SBT). This optimization problem, however, falls into the framework of mixed integer and nonlinear programming (MINLP) [3–5]. Its nondeterministic polynomial hard (NP-hard) nature makes it difficult to find an optimal solution. Traditional tools often require significant computational effort to solve such problems, which are difficult to meet the real-time requirement of practical applications. Deep learning has been adopted to learn the relationship between environmental parameters and the optimized resource allocation. With a well-trained neural network, learning based methods offer the advantage of low inference complexity in decision making [6,7].

A primary challenge in applying learning based methods for MINLP problems is the discrete nature of the policy to be learned. Neural network parameters are typically updated via gradient descent algorithms, while the discretizing process that converts the continuous output of deep neural networks (DNNs) to discrete variables incurs either vanishing or exploding gradients. Four kinds of methods have been proposed to address this issue in the literature. The first method employs the straight-through estimator (STE), which simply sets the gradient of non-differentiable functions used for discretizing process as one during gradient backpropagation [8]. However, while efficient in computation, this method entails gradient mismatch, affecting training performance. The second method relaxes discrete variables into continuous counterparts and then introduces additional constraints to ensure that these variables take only discrete values [4, 9]. This method shifts the challenge from learning discrete policy to ensuring that the learned policy satisfies constraints. As we will see later, making the learned policy satisfy constraint is not a simple task. The third method approximates discretizing process with smoothing functions like the Sigmoid function [10] or projection networks [11]. This method inevitably introduces approximation errors to the original functions, which may degrade the learning performance. Existing methods introduce smoothing parameters to control approximation errors. Typically, these parameters either remain constant throughout training or are adjusted using an annealing approach [10, 12, 13]. For instance, the Gumbel-Softmax reparameterization method employs a smoothing parameter, called temperature, within the SoftMax function, which gradually decreases across training iterations based on an annealing method [12, 13]. Yet, tuning these smoothing parameters in complex problems is a time-consuming task. The fourth method applies the reinforcement learning, such as deep Q-networks (DQN). DQN estimates the value of each discrete action combination, and subsequently selects the optimal action based on its value [14]. However, as the dimensionality of actions increases, the number of action combinations grows exponentially, complicating the learning process. To tackle this obstacle, a branching dueling Q-network (BDQ) was introduced in [15], which achieves a linear growth in the number of action combinations.

Another key challenge lies in ensuring the learned resource allocation policy to satisfy the QoS constraints. Designing an activation function for the output layer of a neural network that satisfies the constraints is a straightforward approach [16]. However, its applicability is largely limited to simple constraints, like ensuring a non-negative power by leveraging activation functions with a non-negative value range. In [17], a projection block is cascaded after a traditional DNN to satisfy complicated rate constraints. Another widely used approach treats constraints as penalties added to the loss function, with weights serving as fine-tuned hyper-parameters [16, 18]. Nonetheless, determining the appropriate penalty weights is demanding, as they affect both the performance and constraint satisfaction. An alternative approach uses Lagrangian dual optimization, where Lagrangian multipliers are learned by the

dedicated *multiplier DNNs* [19,20]. This approach allows for dynamic adjustment of multipliers. However, the constraints cannot be ensured to be satisfied, which is problematic, especially for applications with strict reliability requirements. To alleviate the learning demands on multiplier DNNs, conservative constraints are often considered during training. For instance, increasing reliability requirements during training can reduce the violation of constraints for the learned policy [20].

In this paper we optimize the joint allocation of users' transmit power and the system's RBs for multiuser uplink transmission, where each user has QoS requirements for both LBT and SBT. The objective is to minimize the RB consumption to satisfy the QoS requirements of users, thereby conserving RB resources to accommodate other potential uplink requests. The formulated resource allocation problem is an MINLP problem. We propose a learning based method to solve this discrete optimization while satisfying the stringent QoS requirements. The main contributions are summarized as follows:

- *Hierarchical algorithm for optimal resource allocation:* The studied MINLP resource allocation problem suffers from prohibitively high complexity using exhaustive searching. To solve the problem and provide a performance benchmark for learning based methods, we propose a hierarchical algorithm to find the global optimal solution in the single-user scenario, which is with a drastically reduced computational complexity. We further extend this algorithm to the multiuser scenario.
- *Adaptive smoothing approximation:* When employing deep learning to solve discrete optimization problems, the training of DNNs through gradient backpropagation performs poorly due to vanishing or exploding gradients. To tackle this issue, we develop an adaptive smoothing method to approximate the discretizing process, where the smoothing parameters are dynamically adjusted based on the updated output of DNNs at every iteration. Such a dynamic adjustment helps in controlling approximation errors and gradient magnitudes of the smoothing functions, thereby enhancing the learning performance.
- *Nonlinear penalty for constraint violations:* The learning methods based on Lagrangian dual optimization suffer from slow updates in the trainable parameters of multiplier DNNs when constraints are nearly satisfied. This hinders the effective penalization of slightly violated constraints. This motivates us to design a nonlinear function to intensify penalties for minor constraint violations. By incorporating the nonlinear function with the proposed adaptive smoothing method, the constraints can be satisfied with high reliability.

The remainder of this paper is organized as follows[1]. The system model and problem formulation are described in Section II. The numerical optimization method and the learning based method are presented in Section III and Section IV, respectively. In Section V, we evaluate the performance of the proposed algorithms. Finally, the conclusion is drawn in Section VI.

[1]A part of this work, specifically the numerical algorithm for single-user scenario, was presented in a conference paper [21]. The material in this journal version is substantially extended, including the numerical algorithm for multiuser scenarios, the learning based method, and new simulations results for performance and constraint violation comparisons.

II. System Model and Problem Formulation

A. System Model

Consider the uplink transmission scenario of M users, where each user has both LBT and SBT data demands, as shown in Fig. 1. For example, in a 5G-empowered industry internet of things system, a customer premise equipment (CPE) may concurrently upload surveillance videos and remote control signals [22]. The considered scenario can be straightforwardly simplified when considering only LBT or SBT.

The base station (BS) is equipped with N_r receive antennas and each user is equipped with a single antenna. The transmissions of users are scheduled on different RBs of an orthogonal frequency division multiple access (OFDMA) system to eliminate inter-user interference. Consider that the channel state information of users is available at the BS. The BS optimizes the resource allocation and informs the users of the allocation decisions for transmission.

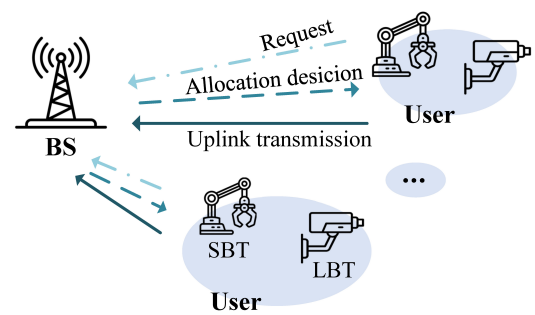

Fig. 1. Uplink transmission with the optimization of resource allocation.

The achievable rate for LBT can be modeled by the Shannon capacity. In contrast, for SBT, the capacity in the short-blocklength regime [23] should be considered to model the achievable rate, which is

$$r_m^S = LB\sum_{f=1}^{F} x_{m,f}^S \log(1+\gamma_{m,f}p_{m,f}) - \frac{1}{\tau}\sqrt{LB\tau\sum\nolimits_{f=1}^{F} x_{m,f}^S\left(1-\frac{1}{(1+\gamma_{m,f}p_{m,f})^2}\right)}Q^{-1}(\epsilon_m), \tag{1}$$

where r_m^S is the achievable rate for SBT of user m, F is the total number of RBs, L is the number of subcarriers in each RB, B is the subcarrier spacing, τ is the duration of a time slot, $x_{m,f}^S \in \{0,1\}$ indicates whether RB_f is assigned to the SBT of user m, $\gamma_{m,f} = \frac{\alpha}{\sigma^2}\|\mathbf{h}_{m,f}\|^2$ represents the noise-normalized effective channel gain of user m in RB_f with maximum ratio combining at the BS (called channel gain for short in the sequel), α is the large-scale channel gain, σ^2 is the noise power, $\mathbf{h}_{m,f} \in \mathbb{C}^{N_r\times 1}$ is the channel vector of user m in RB_f, which is assumed to remain constant across the subcarriers in RB_f over the duration τ, $p_{m,f}$ is the power allocated to RB_f by user m, and $Q^{-1}(\cdot)$ is the inverse of Q function, and ϵ_m is the decoding error probability of user m.

The dispersion $1-\frac{1}{(1+\gamma_{m,f}p_{m,f})^2}$ in (1) is upper bounded by 1. The bound is tight when the channel gain $\gamma_{m,f}$ is large [24], which is often the common case in 5G and beyond systems equipped with large antenna arrays at the BS.

B. Problem Formulation

We optimize the joint allocation of users' transmit powers and the system's RBs to meet the QoS requirements for both LBT and SBT. For user m, the QoS requirement for LBT is characterized by the minimal rate requirement R_m^L, while the QoS requirement for SBT is characterized by both the minimal rate requirement R_m^S and the decoding error probability ϵ_m, as defined in (1). The objective is to minimize the RB consumption, thereby conserving RB resources to accommodate other potential uplink requests.

The optimization problem can be formulated as

$$\min_{\substack{p_{m,f},\\ x_{m,f}^L, x_{m,f}^S}} \sum_{m=1}^{M}\sum_{f=1}^{F}\left(x_{m,f}^L + x_{m,f}^S\right) \tag{2a}$$

$$\text{s.t.} LB\sum_{f=1}^{F} x_{m,f}^L \log(1+\gamma_{m,f}p_{m,f}) \geq R_m^L, m=1,\ldots,M, \tag{2b}$$

$$r_m^S \geq R_m^S, m=1,\ldots,M, \tag{2c}$$

$$\sum_{m=1}^{M}\left(x_{m,f}^L + x_{m,f}^S\right) \leq 1, f=1,\ldots,F, \tag{2d}$$

$$x_{m,f}^L, x_{m,f}^S \in \{0,1\}, m=1,\ldots,M, f=1,\ldots,F, \tag{2e}$$

$$\sum_{f=1}^{F}\left(x_{m,f}^L + x_{m,f}^S\right) p_{m,f} \leq P_{\max}, m=1,\ldots,M, \tag{2f}$$

$$p_{m,f} \geq 0, m=1,\ldots,M, f=1,\ldots,F, \tag{2g}$$

where for user m, the variable $x_{m,f}^L \in \{0,1\}$ indicates whether RB_f is assigned to the LBT, constraints in (2d) and (2e) ensure that RB_f can be assigned to at most one block of a user, constraints (2b) and (2c) define the QoS requirements, (2f) and (2g) are the power constraints of users, and $P_{\max}$ denotes the maximal transmit power of each user.

Due to continuous power allocation and discrete RB allocation indicator, problem (2) is a MINLP problem. Thus, existing convex optimization techniques are not useful here. Its global optimal solution can be obtained by exhaustive searching, which, however, has a prohibitively high complexity. Hence, we resort to deep learning based methods to find a solution. Before delving into the learning methods, in the next section, we first develop two efficient numerical algorithms, which will serve as performance benchmarks for the subsequent learning based methods.

III. Numerical Optimization for Resource Allocation

In this section, we first consider a simplified version of problem (2) involving a single user with both LBT and SBT data demands. We propose a hierarchical optimization algorithm to find the global optimal solution with low complexity. Then, based on this hierarchical method, a heuristic method is developed for multiuser scenario.

A. Global Optimal Solution for Single-user Scenario

We consider the single-user scenario in this subsection, where $M=1$ and the subscript m is omitted in problem (2).

While only a single user is considered, the problem is still complex to solve. We need to determine how many and which RBs should be allocated to LBT or SBT with how much power, aimed at minimizing the total number of occupied RBs, subject to the QoS requirements.

Considering that the LBT and SBT of a single user experience the same channel conditions, an optimal solution to problem (2) should select RBs with large channel gains in order to minimize the total number of required RBs. This suggests a hierarchical algorithm. **In the outer loop**, the RBs are sorted in a descending order based on channel gains, and then the RBs with the highest channel gains are iteratively added into the occupied RBs set $\mathcal{N}$ until the QoS requirements are satisfied. **In the inner loop**, a feasibility problem is solved, determining whether the QoS requirements can be satisfied using the allocated RBs within $\mathcal{N}$. The feasibility problem given $\mathcal{N}$ can be formulated as

$$\min_{x_f^L, x_f^S, p_f} 1 \tag{3a}$$

$$\text{s.t.} \quad \sum_{f\in\mathcal{N}} x_f^L \log(1+\gamma_f p_f) \geq \frac{R^L}{LB}, \tag{3b}$$

$$\sum_{f\in\mathcal{N}} x_f^S \log(1+\gamma_f p_f) - Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} \geq \frac{R^S}{LB}, \tag{3c}$$

$$N^S = \sum_{f\in\mathcal{N}} x_f^S, \tag{3d}$$

$$x_f^L + x_f^S = 1, f\in\mathcal{N}, \tag{3e}$$

$$x_f^L, x_f^S \in \{0,1\}, f\in\mathcal{N}, \tag{3f}$$

$$\sum_{f\in\mathcal{N}} p_f \leq P_{\max}, \tag{3g}$$

$$p_f > 0, f\in\mathcal{N}, \tag{3h}$$

where the constraints come from the constraints of problem (2) by considering that all the RBs in set $\mathcal{N}$ are occupied in the inner loop.

This hierarchical algorithm sequentially increases the number of occupied RBs until all constraints in (2) are satisfied. Consequently, it is able to identify the minimum number of RBs required to satisfy all constraints, thereby achieving global optimality.

The feasibility of problem (3) can be determined by solving the following problem (4), which maximizes the data rate of the LBT while ensuring the QoS requirement of the SBT.

$$\max_{x_f^L, x_f^S, p_f} LB\sum_{f\in\mathcal{N}} x_f^L \log(1+\gamma_f p_f) \tag{4a}$$

$$\text{s.t.} \sum_{f\in\mathcal{N}} x_f^S \log(1+\gamma_f p_f) - Q^{-1}(\epsilon)\sqrt{\frac{N^S}{LB\tau}} = \frac{R^S}{LB}, \tag{4b}$$

$$\sum_{f\in\mathcal{N}} p_f = P_{\max}, \tag{4c}$$

(3d), (3e), (3f), (3h).